

How a clownfish swims

the biomechanics behind the aquarium level's fish rig · *Amphiprion ocellaris* · numbers, formulas and how far to trust each one

The level is ×10 in **space** and real in **time**: speeds are in body lengths per second, frequencies in hertz. The fish is 3.5 in (8.9 cm) real, 0.889 m in the level; cruise 3.048 m/s (3.43 BL/s).

A · Illustrative body wave: authored envelope and wavelength



eight phases of one beat: 0, 1/8 ... 7/8

solid fill: phase 0 · dashed outline: phase 1/4

$$y(s, t) = A(s) \cdot L \cdot \sin[2\pi (s/\lambda - f \cdot t)]$$

$$A(s) = (A/2) \cdot (0.1 + 0.9 s^2), \lambda = L$$

s = 0 at the nose, 1 at the tail tip

ours · envelope shape: drawing only

ours · A = 0.2 L, λ = L: authored

A ≈ 0.2 L
peak to peak

L = body length: 3.5 in (8.9 cm) real · 0.889 m at ×10

swimming direction

wave travels head → tail

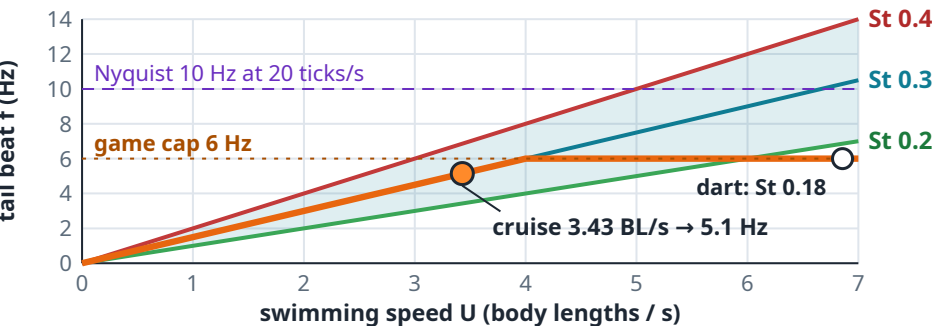
B · Tail-beat frequency follows speed (Strouhal number)

$$f = St \cdot U / A, A = 0.2 L \Rightarrow f = 5 \cdot St \cdot U / L$$

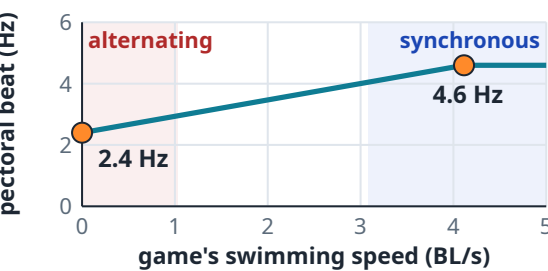
verified · efficient St 0.2–0.4

ours · A = 0.2 L

ours · St = 0.3, cap



D · Pectoral fins: beat rate and gait



verified · A. ocellaris 2.4 → 4.6 /s, surge trials

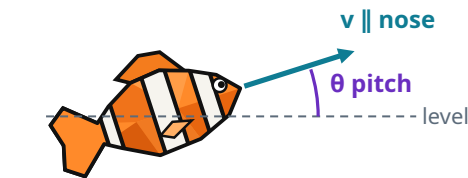
other species · gait switch: hint

slow: alternating, 2.4 Hz

fast: synchronous, 4.6 Hz

0.9 s shown

F · Facing, velocity along the nose, and the control chain



ψ yaw = ROTATION_C
θ pitch = −ROTATION_B, limit 40°
φ roll = ROTATION_A, bank ≤ 15°

the physics hull never turns;
only the visible parts do

ours · every box: the game

verified · layering idea: Tu & Terzopoulos

buttons → wanted direction d

target facing ψ*, θ* (pitch ≤ 40°)

yaw, pitch springs (ω, ζ) · bank

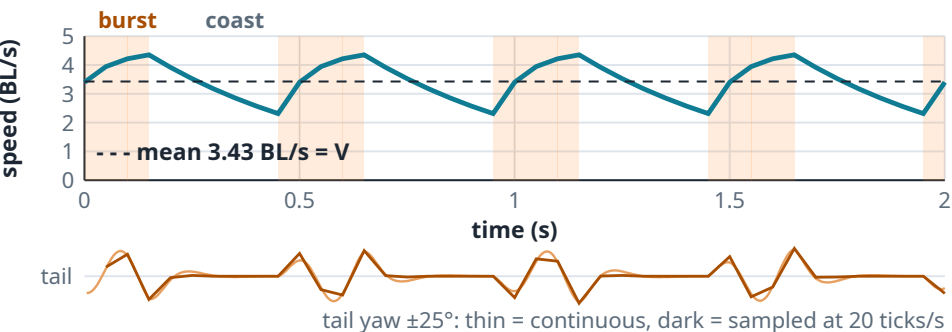
burst-and-coast speed U

$$\mathbf{v} = U \cdot (\cos\theta \cos\psi, \cos\theta \sin\psi, \sin\theta)$$

XSPEED · YSPEED · ZSPEED

C · Burst-and-coast: the gait the game gives the fish

cycle 0.5 s: burst 40 % (tail beats) · coast 60 % (body straight)



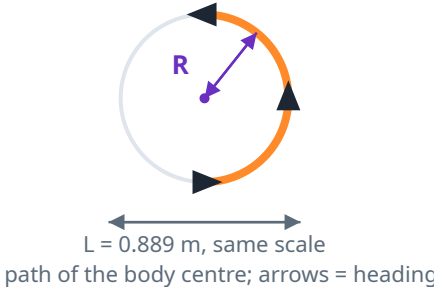
verified · drag ≈ 4 : 1, ≈ 45 % saved (koi)

ours · cycle, share, τ

E · Turning: the head leads, the body bends

cruising U-turn: R ≈ 0.4 m = 0.44 L

$$R = V / (2\pi \cdot \omega), V = 3.048 \text{ m/s}, \omega = 1.25 \text{ rev/s}$$



ours · R: game measurement

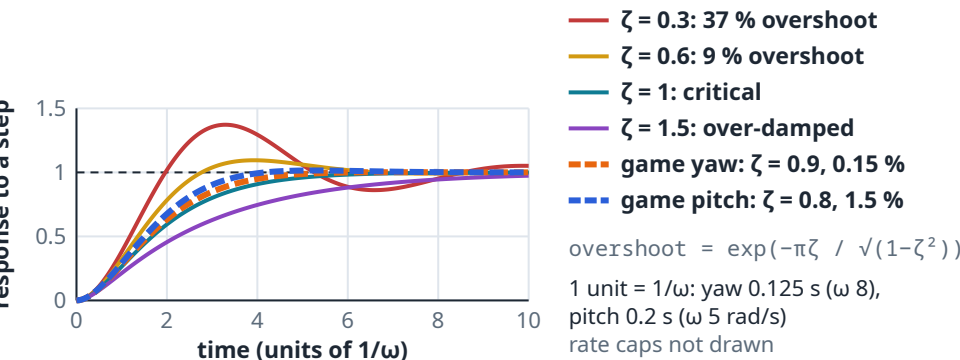
verified · C-start: Domenici & Blake

C-start: before + three stages



stage 1: C-bend; stage 2: propulsion;
stage 3: continued swimming or coasting.
Definitions, not a clownfish figure.

G · Damped steering: why ζ decides how a turn feels



ours · the game's ζ and ω; the other four are illustrations

H · Parameter table: every number, its unit, its source and how far to trust it

Quantity	Value	Chip · source	Quantity	Value	Chip · source	Quantity	Value	Chip · source
Body length L	3.5 in (8.9 cm)	ours	Nearly constant bout red-nose tetras		other species S6	Pitch ω	5 rad/s	ours
Level scale, space	×10	ours	Gait cycle	0.5 s	ours	Pitch ζ	0.8	ours
Cruise speed V	3.048 m/s	ours	Burst share	40 %	ours	Pitch rate cap	0.4 rev/s	ours
Efficient St range	0.2–0.4	verified S2	Burst speed target	3.99 m/s (1.309 V)	ours	Pitch limit	40 °	ours
Strouhal, trout	0.19 → 0.22/0.23	verified S2	Burst rise τ	0.1 s	ours	Pitch limit, diag.	30 °	ours
Strouhal, game	0.3	ours	Coast decay τ	0.5 s	ours	Bank at full yaw	15 °	ours
Tail excursion A/L	0.2 peak to peak	ours	Glide τ (released)	0.45 s	ours	Burst dip in turns	up to 25 %	ours
Wavelength λ/L	≈ 1	ours	Pectoral beat, slow	2.4 Hz	verified S4	Wall easing time	0.25 s	ours
Envelope A(s)	0.1 + 0.9 s²	ours	Pectoral beat, fast	4.6 Hz	verified S4	U-turn radius R	≈ 0.4 m (0.39)	ours
Tail beat at V	5.14 Hz	ours	Speed at fast beat	3.658 m/s (1.2 V)	ours	C-bend gain	0.05 s	ours
Tail-beat cap	6 Hz	ours	Alternating below	0.914 m/s (0.3 V)	other species S5	Body recoil share	20 %	ours
Engine tick rate	20 ticks/s	ours	Synchronous above	2.743 m/s (0.9 V)	other species S5	Tail envelope time	0.06 s	ours
St in a dart	0.18	ours	Damselfish switch	1.87–4.95 BL/s	other species S5	C-start stages	definitions	verified S7
Dart speed	6.096 m/s	ours	Yaw ω	8 rad/s	ours	Layering idea	idea only	verified S8
Drag burst : coast	≈ 4 : 1 (koi)	verified S3	Yaw ζ	0.9	ours			
Energy saved (koi)	≈ 45 %	verified S3	Yaw rate cap	1.25 rev/s	ours			

Sources · checked research; game and diagram choices labelled

S1 Knight 2014, JEB 217:2224
summary; primary results in S2
S2 Nudds et al. 2014, JEB 217:2244
trout best fits: 0.19 → 0.22/0.23
S3 Wu, Yang & Zeng 2007, JEB 210:2181
koi: drag ≈ 4 : 1; ≈ 45 % saved

S4 Marcoux & Korsmeyer 2019, JEB 222(4)
A. ocellaris pectoral 2.4 → 4.6 /s, surge
S5 Hale et al. 2006, JEB 209:3708
damselfish, not clownfish: gait switch
S6 Li et al. 2021, Commun. Biol. 4:40
red-nose tetras: nearly constant bout

S7 Domenici & Blake 1997, fast-starts
C-start stages; our diagram is schematic
S8 Tu & Terzopoulos 1994
architecture only; paper uses muscles

How to read the chips
verified claim checked in the source
ours a game tunable or our own maths
other species another species: a hint only
“ours” values come from the game's code at build time. Not used: Rohr & Fish 2004 (cetaceans). Data dated 2026-09-30. Sources checked 2026-10-06.